

Basic Code Analysis: Git Branching & Merge Conflict Identification

Duration : 30 Min.

Scenario

A pull request failed due to merge conflicts between feature branch 'feature/login' and 'main'. Inspect the Git conflict markers and record conflicting line details in solution.json.

Target File Location & Creation

File to Create/Update: student_workspace/solution.json

Task Objectives

Perform the following actions inside the student_workspace directory: - Create/update 'solution.json' inside 'student_workspace/' with fields 'conflicting_branch' ('feature/login') and 'conflict_file' ('auth.js').

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
2. Initialize the repository history by running: `python setup_git.py`
3. Use Git CLI tools in the terminal to inspect the commit history.
4. Create/update student_workspace/solution.json and record the extracted commit hash and author username.
5. Save your changes (Ctrl + S or Cmd + S).
6. Verify your work by running `python run.py` in the terminal before submitting.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	solution.json exists in student_workspace/	3 Marks

TC2	Conflicting branch name ('feature/login') identified	4 Marks
TC3	Conflict marker file ('auth.js') correctly identified	3 Marks

Total Score: 10 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.